WORKED SOLUTION

STEP III 1994, Question 6

Pure • Rotations in the Argand Diagram

STEP III 1994, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q6

The key idea

A rotation about the point p through angle θ is the map $z \mapsto p + e^{i\theta}(z - p)$: shift p to the origin, rotate, shift back. Composing several such maps always gives something of the form $z \mapsto \lambda z + \mu$, and "$P_{\text{last}} = P_1$ for *every* P_1" means this map is the identity, so $\lambda = 1$ and $\mu = 0$. Everything else is reading those two equations geometrically.

Solution

Finding z_5

A positive quarter-turn is multiplication by i, so rotating P_1 about A gives

$$z_2 = a + \mathrm{i}(z_1 - a) = (1 - \mathrm{i})a + \mathrm{i}z_1.$$

In the same way $z_3 = (1 - \mathrm{i})b + \mathrm{i}z_2$, $z_4 = (1 - \mathrm{i})c + \mathrm{i}z_3$ and $z_5 = (1 - \mathrm{i})d + \mathrm{i}z_4$. Substituting back step by step,

$$\begin{aligned} z_5 &= (1 - \mathrm{i})(d + \mathrm{i}c + \mathrm{i}^2 b + \mathrm{i}^3 a) + \mathrm{i}^4 z_1 \\ &= (1 - \mathrm{i})(d - b + \mathrm{i}(c - a)) + z_1. \end{aligned}$$

When does P_5 return to P_1?

Four quarter-turns make a full turn, so the coefficient of z_1 is exactly 1, and

$$z_5 - z_1 = (1 - \mathrm{i})(d - b + \mathrm{i}(c - a)),$$

which does not depend on z_1 at all. The composite map is a pure translation. Since $1 - \mathrm{i} \neq 0$,

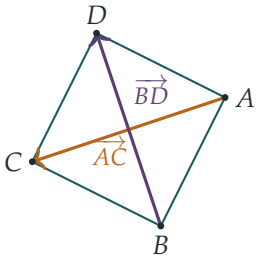
$$\begin{aligned} z_5 = z_1 \text{ for every } z_1 &\iff d - b + \mathrm{i}(c - a) = 0 \\ &\iff \mathrm{i}(c - a) = b - d \\ &\iff a - c = \mathrm{i}(b - d), \end{aligned}$$

where the last step multiplies both sides by $-\mathrm{i}$ (and $-\mathrm{i}\cdot\mathrm{i}=1$). Every step is reversible, so this is the required "if and only if". (If the condition fails, the translation is non-zero, and then $P_5 \neq P_1$ for *every* choice of P_1.)

Ask yourself What does $a-c=\mathrm{i}(b-d)$ say about the shape $ABCD$?

Multiply by -1: the condition is $c-a=\mathrm{i}(d-b)$. Now $c-a$ represents the vector $\overrightarrow{AC}$ and $d-b$ the vector $\overrightarrow{BD}$. Multiplying by i rotates a vector by $\pi/2$ without changing its length. So:

$\overrightarrow{AC}$ is $\overrightarrow{BD}$ turned through a positive right angle; that is, the diagonals AC and BD of the quadrilateral $ABCD$ are equal in length and perpendicular, with AC a quarter-turn anticlockwise from BD.



In the example drawn, $\overrightarrow{BD}=-1+3\mathrm{i}$ and $\overrightarrow{AC}=-3-\mathrm{i}=\mathrm{i}(-1+3\mathrm{i})$. The orientation matters: a square labelled *clockwise* satisfies the condition, but one labelled anticlockwise does not (there $\overrightarrow{AC}=-\mathrm{i}\,\overrightarrow{BD}$).

Three rotations through θ

Write $w=\mathrm{e}^{\mathrm{i}\theta}$. Exactly as before, each step is $z\mapsto(1-w)p+wz$, so

$$\begin{aligned} z_4 &= (1-w)c+w\big((1-w)b+w\big((1-w)a+wz_1\big)\big)\\ &= (1-w)\big(c+wb+w^2a\big)+w^3z_1. \end{aligned}$$

This is $\lambda z_1+\mu$ with $\lambda=w^3$ and $\mu=(1-w)(c+wb+w^2a)$. It equals z_1 for every z_1 if and only if $\lambda=1$ and $\mu=0$ (take $z_1=0$ to get $\mu=0$, then $z_1=1$ to get $\lambda=1$).

The angle. $w^3=1$ means 3θ is a multiple of 2π. We read $\theta\neq 0$ as "not a whole number of turns", so $w\neq 1$; hence w is a non-real cube root of unity and

$$\theta=\pm\tfrac{2\pi}{3}\pmod{2\pi}.$$

The points. Since $w\neq 1$, $\mu=0$ becomes $c+wb+w^2a=0$. Multiplying by w and using $w^3=1$ gives $a+w^2b+wc=0$, and since $1+w+w^2=0$ (as $w^3=1$, $w\neq 1$) we can rewrite $a=-(w+w^2)a$, so

$$w(c-a)+w^2(b-a)=0 \quad\Longrightarrow\quad c-a=-w\,(b-a)=\mathrm{e}^{\mathrm{i}(\theta+\pi)}(b-a)=\mathrm{e}^{\mathrm{i}(\theta-\pi)}(b-a).$$

Ask yourself What is $e^{i(\theta-\pi)}$ when $\theta = \pm 2\pi/3$?

For $\theta = 2\pi/3$ it is $e^{-i\pi/3}$; for $\theta = -2\pi/3$ it is $e^{i\pi/3}$ (since $e^{-5\pi i/3} = e^{i\pi/3}$). So $\overrightarrow{AC}$ is $\overrightarrow{AB}$ turned through $\mp\pi/3$ with the same length: ABC is **equilateral**, with A, B, C in **clockwise** order when $\theta = 2\pi/3$ and **anticlockwise** order when $\theta = -2\pi/3$. Conversely, each of these steps reverses, so these conditions also give $\mu = 0$.

Since A, B, C lie on the unit circle, that circle is the triangle's circumcircle, and for an equilateral triangle the circumcentre is the centre, so the vertices are equally spaced around $|z| = 1$. Both cases combine neatly into one statement:

$$\theta = \pm\tfrac{2\pi}{3} \pmod{2\pi}, \qquad b = a\,e^{-i\theta}, \quad c = b\,e^{-i\theta}.$$

In words, the three rotations are by a third of a turn, and A, B, C are equally spaced on the unit circle, stepping round in the *opposite* sense to the rotation.

Checks

- A computer applied the four quarter-turns to random points about random A, B, C, D and matched the formula for z_5 to 10^{-9}; with $c = a - i(b - d)$ every P_1 returned home, and nudging c by 0.1 broke it.
- For random a on the unit circle and $\theta = \pm 2\pi/3$, $b = ae^{-i\theta}$, $c = be^{-i\theta}$ gave $z_4 = z_1$ for random z_1; reversed labelling and other θ did not.
- By hand: rotate $z_1 = 0$ about the clockwise unit square $a = 0$, $b = i$, $c = 1 + i$, $d = 1$. You get $0 \to 0 \to 1 + i \to 1 + i \to 0$. Back home.

Common slips

- Writing a rotation about A as $z \mapsto iz$, forgetting to shift A to the origin first.
- Dropping orientation: the diagonals' sense of rotation, and the triangle's for each of $\theta = \pm 2\pi/3$, are part of the conditions.

Beyond STEP

The maps $z \mapsto \lambda z + \mu$ with $|\lambda| = 1$ are the orientation-preserving isometries of the plane, and they form a group. Angles add, so a composite of rotations is a rotation by the total angle, or a translation when the total is a whole number of turns: that is why $z_5 - z_1$ came out constant. Needham's *Visual Complex Analysis* develops this geometric view of complex numbers beautifully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.